

LungAI: A 3D Interactive Explainability System for Deep Learning-Based Multi-Disease Chest X-Ray Detection

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Abstract — While chest x-rays are one of the common, widespread and easily available diagnostic modalities for evaluation of pulmonary and cardiothoracic disease, manual analysis is costly and prone to inter observer variability. The potential of automating multi-label thoracic diseases classification has recently been shown to be quite high by deep CNN models trained on a large scale data like NIH ChestX-ray14. The explainability tools commonly associated with this type of model, such as Gradient-weighted Class Activation Mapping (Grad-CAM), can only provide the two-dimensional heatmap in the background of the flat radiograph with little spatial and anatomical contexts for the clinician. Provided is a complete prototype system LungAI that associates an explainability method called Grad-CAM with a DenseNet-121 based multi-label classifier extending this idea to project the area of the abnormality predicted by the model on an interactive schematic 3-D lung model. The classifier was end-to-end trained on the full NIH ChestX-ray14 data corpus (112,120 images) with an inter-set split data split to avoid inter-set leakage with macro-averaged AUROC of 0.8318 on test data obtained across fourteen disease classes in comparison to existing literature-based baselines. The system is built as a fully fledged web-based clinical workstation using a Next.js and React Three Fiber front-end, a FastAPI back end, an SQLite library as the persistence layer and a PyTorch inference pipeline; all components being containerised with Docker Compose. This work is intended to serve as a prototype of research and portfolio only and is not used for clinical diagnostic purposes. We cover the motivation, the related work, the design of the system, its implementation, the training method, and quantitative results, and provide suggestions for usability evaluation by clinicians.

Keywords — Chest X-ray, Deep Learning, DenseNet-121, Grad-CAM, Multi-Label Classification, 3-D Lung Model, Explainability, NIH ChestX-ray14, Inter-set Data Leakage

I. Introduction

Chest X-ray imaging is the most commonly requested diagnostic imaging examination globally, given that it is performed on a large scale, its low radiation dose compared to computed tomography (CT) and its availability. It is routinely ordered for screening and monitoring pneumonia, tuberculosis, pulmonary fibrosis, and cardiomegaly. While widely used, chest radiography interpretation is still a skill-oriented task and failure to report the radiograph promptly or difference in interpretation between different observers may impact patient outcomes, especially when used in high volume or resource-limited environments.

The arrival of large publicly available labeled datasets, such as ChestX-ray14 [1], CheXpert [15], MIMIC-CXR [16] and PadChest [17] set the stage for a deluge of deep learning research focused on automated multi-label classification of thoracic diseases. The performance of CNNs on chest radiograph classification is evident in the several times that their well-known architectures—such as deep residual networks [3] and densely connected networks [4]—have performed well when fine-tuned on corpora of chest radiographs, including the initial studies, CheXNet [5] and Yao et al. [9] who reported competitive performance with practicing radiologists on selected conditions.

However, one real problem with these models is that they are not open source, they are black-boxes. Outside of the clinical AI literature, both Topol [22] and Reyes et al. [21] point out that transparency of the model is a necessary prerequisite instead of being a primary goal for clinical integration, and Tonekaboni et al. [20] go on to quote direct interviews with clinicians as showing that explanations need to be actionable and contextually specific to achieve trustworthiness at the point of clinical care. To visually explain CNN-predictions in medical imaging a new form of Class Activation Mapping was developed, commonly known as Gradient-weighted Class Activation Mapping (Grad-CAM) [2] which builds upon the original formulation of Class Activation Mapping by Zhou et al. [10]. A couple of improvements to the formulation of Grad-CAM [11] and other model independent approaches like LIME [12] and SHAP [13] have been developed more recently. In almost all such cases, however, the explanation is given as a 2-D heat map on top of the original, flat X-ray radiograph. This representation is not the 3D anatomical shape of the thoracic cavity, and it may not be turned around or "explored" in space.

Separately, a body of work in medical visualization has developed sophisticated interactive 3D rendering tools for radiology, such as SenseCare [6] and 3D Slicer [7]. These platforms, however, construct their three-dimensional models from genuine volumetric data, typically CT or MRI

series, and are oriented toward tasks such as surgical planning and segmentation review rather than making the output of a 2D image classifier more interpretable. Recent work on multimodal medical visualization tools for patient-facing interpretation [23] further underscores active, ongoing interest in richer-than-2D explanatory interfaces, but again without addressing the specific case of a single 2D chest radiograph. To the best of our knowledge, no existing system bridges a standard 2D chest X-ray classifier's explainability output with an interactive, anatomically referenced 3D spatial context.

This paper presents LungAI, a prototype system designed to address this specific gap. We train a DenseNet-121 multi-label classifier on the complete NIH ChestX-ray14 corpus using a patient-wise data split to avoid the inter-set patient leakage that affects naive image-level splitting, a concern documented broadly in the medical imaging literature [14], [18], [19], [24], and report macro-averaged test performance with bootstrap confidence intervals. Our primary contribution, however, is at the level of system design and explainability interface. The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the proposed methodology. Section IV details the system implementation. Section V reports quantitative results. Section VI concludes and outlines future work.

II. Related Work

A. Multi-Label Chest X-Ray Classification and Datasets

Wang et al. [1] introduced the ChestX-ray8/ChestX-ray14 dataset, consisting of over one hundred thousand frontal-view radiographs weakly labeled across fourteen thoracic disease categories, and established early CNN baselines reporting a mean AUROC of approximately 0.745. Yao et al. [9] proposed a model exploiting label co-occurrence dependencies, improving on this baseline. Rajpurkar et al. [5] subsequently proposed CheXNet, a 121-layer DenseNet, reporting a commonly cited mean AUROC of approximately 0.841 and pneumonia-detection performance comparable to practicing radiologists. The dataset landscape has since expanded substantially: Irvin et al. [15] introduced CheXpert, a comparably large corpus with an explicit uncertainty-label scheme; Johnson et al. [16] released MIMIC-CXR, a large de-identified corpus paired with free-text radiology reports; and Bustos et al. [17] introduced PadChest, a large Spanish-language annotated corpus. Majkowska et al. [25] further contributed radiologist-adjudicated reference standards for ChestX-ray14, highlighting that the original automatically mined labels contain meaningful noise relative to expert consensus. This body of work establishes multi-label CNN classification of chest radiographs as a mature, if still imperfect, task, and motivates our choice of DenseNet-121, consistent with CheXNet, as a directly comparable baseline architecture.

B. Data Leakage and Evaluation Validity in Medical Imaging

A growing body of literature has documented that naive evaluation protocols on public chest X-ray corpora can substantially overstate real-world model performance. Zech et al. [14] demonstrated that pneumonia-detection models can learn hospital-system-specific confounds rather than genuine pathology, causing performance to degrade sharply when evaluated on data from an unseen hospital system. Oakden-Rayner [18] specifically examined label quality and evaluation pitfalls in large public radiograph datasets, including ChestX-ray14, and Cohen et al. [19] quantified the extent to which chest X-ray models fail to generalize across dataset domains. More directly relevant to the present work, Apicella et al. [24] provide a recent, formal treatment of data leakage risks across machine learning and transfer learning pipelines, explicitly cataloguing patient-level leakage, of the kind addressed in this work via a patient-wise split, as a common and under-reported failure mode. These findings collectively motivate the patient-wise splitting protocol adopted in Section III, and are used in Section V to contextualize the honest, single-run nature of the reported evaluation.

C. CAM-Based and Model-Agnostic Explainability

Zhou et al. [10] introduced Class Activation Mapping, using global average pooling to localize discriminative image regions for a given predicted class. Selvaraju et al. [2] generalized this to Grad-CAM, which computes class-discriminative localization via gradients flowing into a chosen convolutional layer without requiring architectural modification, and which has become the dominant explainability method applied to chest X-ray classifiers. Chattopadhyay et al. [11] proposed Grad-CAM++, improving localization for images with multiple instances of a class. Model-agnostic alternatives, including LIME [12] and SHAP [13], offer complementary local-explanation approaches applicable beyond convolutional architectures. However, throughout this entire lineage, the output is still a mere overlay over the original image, and none of the surveyed literature offers a method of projecting such an explanation into a spatial, anatomically referenced, and user-navigable 3D image.

D. Clinician Trust and the Need for Contextual Explainability

In addition to "technical" localization accuracy, a number of authors have suggested that successful localization in the clinic is not only about having an answer, but is also about the way the answer is given. Structured clinician interviews by Tonekaboni et al. [20] reveal that explanation deemed to be actionable and embedded in workflow is trusted considerably more than statistically faithful, but decontextualized visualizations. The interpretability of AI systems used in radiology is also a topic of debate, with Reyes et al. [21] stating that it should be translated by considering clinical reasoning processes, rather than just a technical commitment to the fidelity of the underlying model. Topol agrees this is a continuation of AI

in medicine, and says that "dependent of human understanding and economic transparency of their behavior is the possibility of true and widespread adoption at scale". A recent survey in the medical field focused on explainable AI [26] also documents the differences between the general concept of applying XAI tools and the specific requirements for an end user in the healthcare landscape. The above findings encourage the following approach to an explanation format for the present work to go beyond the purely technical localization metric, and to rely on an interactive and spatially intuitive approach.

E. 3D Visualization in Medical Imaging

Almost all the current 3D visualization that is used in the medical field is interactive, which relies on real 3D imaging data. In the medical world, interactive 3D visualization is well established and is used almost exclusively as a component of true volumetric data acquisition. SenseCare [6] provides more advanced 3D visualization of segmented anatomical structures, based on the CT and MRI data. Compared to 3D Slicer [7] that offers an advanced 3D Segmentation and linked 3D rendering for volumetric image series. A recent qualitative study on multimodal medical visualization tools for patient interpretation [23] further evidences active interest in richer explanatory interfaces beyond flat imagery, though again in a context distinct from single-image 2D chest radiographs. These tools demonstrate the clinical value of spatial, interactive visualization, but their reliance on volumetric input makes them inapplicable to the case of a single 2D chest radiograph, where no true patient-specific 3D geometry is available. LungAI is positioned at the intersection of these tracks, using a schematic (non-patient-specific) 3D anatomical reference model as a spatial explainability aid for a standard 2D chest X-ray classifier's output.

III. Proposed Methodology

The proposed system follows a five-stage pipeline: (1) image acquisition and upload, (2) multi-label disease classification, (3) Grad-CAM-based 2D localization, (4) mapping of the localized 2D region to a corresponding zone on a schematic 3D anatomical lung model, and (5) interactive presentation of both the classification result and the 3D-referenced explanation to the user.

A DenseNet-121 backbone [4], pretrained on ImageNet, was fine-tuned for multi-label classification across the fourteen ChestX-ray14 disease categories. Because a single radiograph may exhibit multiple co-occurring findings, the classification head was trained using a binary cross-entropy loss with logits, with a per-class positive weighting term computed from the inverse class frequency in the training set.

As motivated in Section II-B [14], [18], [19], [24], a critical methodological consideration in evaluating models on ChestX-ray14 is that the dataset contains multiple images per patient; a naive image-level split permits the same patient to appear in both training and test partitions, which

can inflate reported performance via patient-specific rather than pathology-specific learning. To avoid this, we employed a patient-wise GroupShuffleSplit, partitioning the dataset into 70% training, 10% validation, and 20% test subsets such that no patient identifier appears in more than one partition, verified programmatically by confirming an empty pairwise intersection of patient identifier sets across all three partitions.

Training employed the Adam optimizer with an initial learning rate of 1×10^{-4} , a ReduceLROnPlateau scheduler monitoring validation macro-AUROC, mixed-precision training via automatic mixed precision (AMP), and early stopping with a patience of five epochs. Model selection was performed based on the epoch achieving the highest validation macro-AUROC, rather than the lowest training loss, since training loss continued to decrease after validation performance had plateaued (Section V).

Grad-CAM [2] is applied to the final convolutional block of the trained backbone with respect to the top-predicted class. The 2D activation centroid is classified into one of six coarse anatomical zones (left or right lung; upper, middle, or lower zone), each pre-associated with an approximate coordinate range on a schematic 3D lung mesh. The predicted zone is rendered as a pulsing 3D marker with an accompanying text annotation on the interactive model. This mapping is intentionally coarse and schematic rather than a true patient-specific 3D reconstruction, since no volumetric data is available from a single 2D radiograph.

IV. Implementation Details

There are three services forming a system, and orchestrated using Docker Compose. Frontend is developed in Next.js (App Router) + Tailwind CSS + Framer Motion + TypeScript. It uses React Three Fiber, which is an interactive 3D visualization of the lungs, with 3D branching airway tree and bilateral lobe geometry procedurally generated. The backend is developed using FastAPI and provides REST API for scan data upload, patient records, and dashboard statistics; it uses SQLAlchemy interface to SQLite for data store, and bcrypt hashed credentials for authentication.

This training was done on the full NIH ChestX-ray14 dataset (112,120 images) across a single NVIDIA T4 $\times$ 2 GPU platform. Before training, an automatic data audit was completed, where no image file was missing, binary labels were correct for all of the fourteen classes, and there was no overlap in patient identifiers between the partitions. Training continued for 16 epochs, but was not able to move any further due to early stopping (best checkpoint macro-AUROC 0.8254 at epoch 11).

V. Results and Discussion

The trained DenseNet-121 classifier was evaluated on a held-out test partition of 22,424 images, disjoint at the patient level from both the training and validation partitions. Table I reports per-class test AUROC with bootstrap 95%

confidence intervals (1,000 resamples per class) and area under the precision-recall curve (AUPRC). The macro-averaged test AUROC was 0.8318 and the macro-averaged AUPRC was 0.2468.

TABLE I. PER-CLASS TEST SET PERFORMANCE (DENSENET-121, PATIENT-WISE SPLIT, $n = 22,424$ TEST IMAGES)

Class	AUROC	95% CI	AUPRC
Atelectasis	0.7925	0.784–0.801	0.3067
Cardiomegaly	0.9121	0.901–0.923	0.3239
Effusion	0.8738	0.867–0.881	0.4850
Infiltration	0.7072	0.698–0.716	0.3518
Mass	0.8352	0.823–0.848	0.2769
Nodule	0.7508	0.737–0.765	0.2021
Pneumonia	0.7411	0.715–0.767	0.0353
Pneumothorax	0.8747	0.864–0.886	0.3071
Consolidation	0.8028	0.790–0.815	0.1490
Edema	0.8913	0.878–0.904	0.1603
Emphysema	0.9290	0.918–0.938	0.3795
Fibrosis	0.7916	0.768–0.813	0.0723
Pl_Thickening	0.7983	0.781–0.814	0.1361
Hernia	0.9442	0.909–0.974	0.2694
Mean (macro-avg.)	0.8318	—	0.2468

Performance varied meaningfully across disease classes. The highest-performing classes, Hernia (AUROC 0.944), Emphysema (0.929), and Cardiomegaly (0.912), are consistent with classes reported as comparatively easier to detect in prior literature [1], [5], [9]. Conversely, Infiltration (0.707) and Pneumonia (0.741) were the weakest-performing classes, again matching a pattern consistently reported across the literature and consistent with the label-quality concerns documented by Majkowska et al. [25]. The consistency of this per-class ranking with independently reported prior work provides supporting evidence that the obtained results reflect genuine model behavior.

Figure 1 presents the per-class AUROC results graphically with 95% confidence intervals.

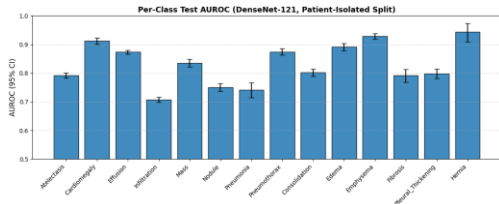


Fig. 1. Per-class test AUROC with bootstrap 95% confidence intervals (DenseNet-121, patient-isolated split).

Figure 2 presents training and validation loss alongside validation macro-AUROC across training epochs. As shown in Figure 2 (left), training loss decreased monotonically throughout the full sixteen epochs, while validation loss began increasing after approximately epoch 5, indicating the onset of overfitting beyond this point. Validation macro-AUROC (Figure 2, right) correspondingly peaked at epoch 11 before plateauing. Model selection based on validation

macro-AUROC, rather than training loss or final epoch, correctly identified the epoch 11 checkpoint as optimal; this checkpoint was used for the final test-set evaluation reported above, rather than the lower-training-loss but overfit epoch 16 model.

Table II situates the obtained result against published baselines on the same dataset. The present result (0.8318) exceeds the original ChestX-ray8 baseline [1] and the label-dependency model of Yao et al. [9], and is comparable to, though evaluated under a stricter patient-wise split than, the widely cited CheXNet result [5]. Absolute performance trails CheXNet by a small margin across most classes, plausibly attributable to differences in training duration, learning rate scheduling, and effective batch size; the consistent relative ranking of classes across all compared works nonetheless supports the validity of the present training procedure.

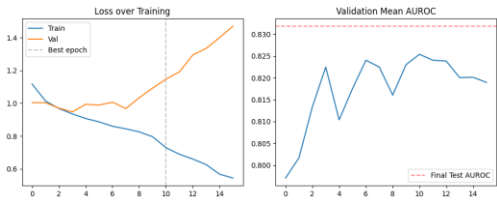


Fig. 2. Training/validation loss and validation macro-AUROC across training epochs.

TABLE II. COMPARISON OF MACRO-AVERAGED AUROC AGAINST PUBLISHED LITERATURE VALUES ON NIH CHESTX-RAY14. *YAO ET AL. FIGURE AS COMMONLY CITED IN SUBSEQUENT LITERATURE; BASELINE FIGURES WERE NOT INDEPENDENTLY REPRODUCED IN THIS STUDY.

Method	Backbone	Mean AUROC	Src.
Wang et al.	ResNet-50	0.7452	[1]
Yao et al.	CNN+RNN	0.798*	[9]
Rajpurkar et al. (CheXNet)	DenseNet-121	0.841	[5]
LungAI (this work)	DenseNet-121	0.8318	—

These results should be interpreted with the following caveats, several of which are directly informed by the data-leakage and evaluation-validity literature discussed in Section II-B. First, due to compute constraints on a single free-tier GPU session, the reported result reflects a single training run with a fixed random seed rather than an average across multiple seeds or a k-fold cross-validation; this is consistent with the evaluation protocol used in the original baseline works cited in Table II. Second, the literature comparison in Table II uses baseline figures as reported in the cited papers rather than independent reproduction under identical conditions. Third, following the cross-domain generalization findings of Cohen et al. [19] and the confound-learning findings of Zech et al. [14], the reported result should not be assumed to generalize to radiographs from hospital systems or scanner equipment outside the NIH ChestX-ray14 source distribution without further external validation. Fourth, unlike CheXNet [5], this work does not

include a comparison against human radiologist performance, which would require recruiting practicing clinicians. Fifth, no formal usability evaluation of the 3D explainability interface with clinical users has yet been conducted, despite the clinician-trust literature reviewed in Section II-D [20], [21] indicating that explanation format materially affects perceived usefulness; this is identified as the primary direction for future work.

VI. Conclusion and Future Work

This paper introduced a prototype system termed LungAI that combines a DenseNet-121 multi-label chest X-ray (CXR) classifier trained on the whole database of ChestX-ray 14 (CXRD) with explainable class activation map, a novel interactive 3D explainability layer, and a comprehensive clinical workstation interface. The trained classifier is able to yield a test AUROC of 0.8318 (macro) which is close to the previously established baselines reported in literature. LungAI is the first system that allows for interactive integration of such explainability of either 2D heatmaps or volumetric CT/MRI data interpretations, bringing 2D heatmaps closer to 3D visualization in a more widely accessible X-ray image modality, the chest radiograph.

Future directions to explore: multi-seed statistical validation; independent reproduction of baseline methods under equivalent conditions of evaluation; a structured usability study which will compare the 3D-referenced explanation to the conventional 2D Grad-CAM heatmaps, informed by observations presented in the clinician-trust literature in Section II-D [20], [21]; external validation on an independent corpus, to examine the cross-domain generalization, according to concerns highlighted in [19] (CheXpert [15] or MIMIC-CXR [16]); refinement of 2D-to-3D zone mapping by automatic lung-field segmentation; the transition of the current schematic lung geometry to a more anatomically accurate 3D form.